

Capacity, Heat Pumps and the Limits of the Inspection

Why bigger is worse, why a heat pump is sized for summer, and the two things you are not required to do.

An air conditioner does two jobs: it lowers the temperature and it removes moisture. Removing moisture takes time, because the air has to stay in contact with the cold coil long enough to condense on it.

That is why an oversized unit is less likely to dehumidify the air. It reaches the thermostat setting quickly and shuts off, so the coil never runs long enough to wring the air out. The house ends up cold and clammy, and the short cycling wears the compressor. Bigger is not better.

Heat pumps

Heat pump	An air conditioner that can run backwards. In winter it takes heat from outside air and moves it indoors.
Heat source	Outside air, for an air-to-air heat pump. Even cold air contains heat that can be moved.
Sized on	The cooling load. It is an air conditioner first, and supplemental electric heat covers whatever the winter needs beyond its capacity.

YOU HAVE MISSED THIS 2 TIMES

When the temperature is below 20 degrees, a heat pump:

Will need auxiliary heaters to supplement the heat.

Below about 20 degrees there is too little heat left in the outside air for the machine to move economically, so auxiliary resistance heaters take over. Sizing it for the heating load instead would leave it oversized all summer, short-cycling and never dehumidifying — the fault described above.

The numbers you test against

The numbers here

Air leaving the register — cooler than room temperature by about	15 degrees
Outdoor air temperature needed before testing a cooling system	60 to 65 degrees F
Refrigerator temperature	35 degrees F

YOU HAVE MISSED THIS 2 TIMES

For A/C, the temperature of the air leaving the register should be about ____ degrees cooler than the room temperature.

15

The outdoor limit is a protection rule, not a convenience. Running a compressor when it is too cold outside can damage it, so if the weather is below the threshold you report that the system could not be tested and say why.

What you are not required to do

YOU HAVE MISSED THIS

Regarding air conditioning, home inspectors are not required to:

Inspect the cooling capacity.

Capacity is a calculation involving the house's heat gain, not an observation. You confirm the system runs and produces a reasonable temperature drop. You do not certify that it is the right size for the building.

Say it back

Cover the page. Answer each one out loud.

1. Name the two jobs an air conditioner does.
2. Why is an oversized unit worse at the second job?
3. What is the heat source for an air-to-air heat pump?
4. What load is a heat pump sized on, and why not the other one?
5. What happens below 20 degrees?
6. Expected temperature drop at the register, and the outdoor limit for testing.
7. What are you not required to inspect on a cooling system?